

F. Build XOR on a Segment

time limit per test: 5 seconds

memory limit per test: 512 megabytes

You are given an array of n integers $a_1, a_2, \dots, a_n$, where all numbers are from 1 to $2^{12} - 1$.

You have to process q queries. Each query is defined by three integers l_i, r_i, x_i : you need to find the smallest set $S = \{s_1, s_2, \dots, s_k\}$ that satisfies the following conditions:

- each s_j is equal to some element from the subarray from the l_i -th position to the r_i -th position inclusive;
- $s_1 \oplus s_2 \oplus \dots \oplus s_k = x_i$, where $\oplus$ denotes bitwise XOR.

Input

The first line contains one integer n ($2 \leq n \leq 10^4$).

The second line contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 2^{12} - 1$).

The third line contains one integer q ($1 \leq q \leq 10^6$).

Then q lines follow. The i -th of them contains three integers l_i, r_i, x_i ($1 \leq l_i \leq r_i \leq n$; $1 \leq x_i \leq 2^{12} - 1$).

Output

For each query, print one integer — the minimum size of the required set. If such a set does not exist, print 0.

Example

input	Copy
7	
3 5 4 1 7 3 1	
5	
1 3 1	
1 4 1	
1 3 2	
4 7 5	
1 7 8	
output	Copy
2 1 3 3 0	

Note

Consider the queries from the example:

- in the first query, you can choose $S = \{5, 4\}$;
- in the second query, you can choose $S = \{1\}$;
- in the third query, you can choose $S = \{4, 3, 5\}$;
- in the fourth query, you can choose $S = \{1, 3, 7\}$.

Educational Codeforces Round 185 (Rated for Div. 2)

Contest is running

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Contestant



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Language: GNU G++17 7.3.0

 Choose file: No file chosen

